

Decoupled Feedforward Control for an Air-Conditioning and Refrigeration System

Neera Jain, *Member, IEEE*, Richard J. Otten, *Member, IEEE*, and Andrew G. Alleyne, *Senior Member, IEEE*

Abstract—Multiple control objectives must be met in order to satisfy the system capacity and efficiency requirements of air-conditioning and refrigeration (A/C & R) systems. Moreover, in the HVAC industry, it is generally preferred to tune multiple single-input-single-output (SISO) control loops rather than a single multiple-input-multiple-output (MIMO) control loop. This paper demonstrates that a SISO decentralized control design, presented in [1], can be enhanced with feedforward control on select control channels to improve the response to reference signals and reduce the effect of system coupling. A feedforward controller is combined with a PI feedback controller to improve the reference tracking performance of average evaporator temperature while reducing the coupling effect of changes in other input channels. Both simulation and experimental results are presented.

I. INTRODUCTION

AN idealized air-conditioning and refrigeration (A/C & R) system, as shown in Fig. 1, is a thermodynamic system driven by the phase characteristics of the refrigerant that is flowing through it. It assumes isentropic compression, isenthalpic expansion, and isobaric evaporation and condensation.

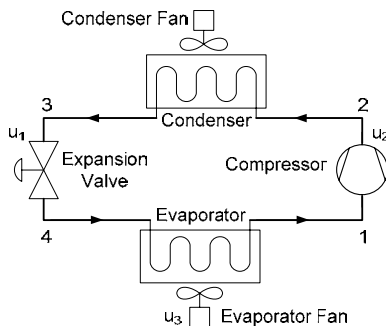


Fig. 1. Schematic of ideal subcritical VCC system.

The basic control objectives of an A/C & R system are to meet desired cooling capacity while maximizing system efficiency. Numerous control schemes have been developed for A/C & R systems with superheat and evaporation temperature (or pressure) as the controlled variables [2] [3][4]. However, these references frequently noted the difficulty of controlling evaporator superheat and temperature (or pressure) with individual single-input-

single-output (SISO) control loops due to the physical coupling between these two variables. Multi-input-multi-output (MIMO) control techniques have been shown to effectively mitigate the I/O coupling [2][3][5]. Nevertheless, industrial practitioners and service engineers in the HVAC industry generally prefer to tune multiple SISO control loops rather than design a more complex MIMO controller to achieve the same objectives.

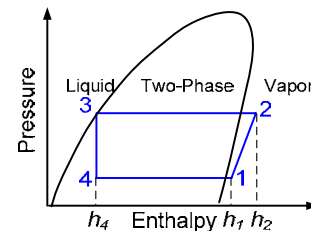


Fig. 2. P-h diagram for ideal subcritical VCC.

This motivated the development of a decoupled control framework for A/C & R systems. In [1] a novel choice of output control variables was shown to effectively decouple system dynamics such that a decentralized control approach, consisting of individual SISO control loops, can be used to meet desired performance objectives. An extensive set of possible input-output (I/O) pairings was considered, and the relative gain array (RGA) technique [6] was used to quantify the reduction in coupling for a given set of I/O pairings over another. The decoupled set of I/O pairings is given in Table

I. Note that $P_{3-4} \triangleq P_3 - P_4$ and $P_{(3+4)/2} \triangleq \frac{1}{2}(P_3 + P_4)$ (and similarly for $T_{(1+4)/2}$) where 3 and 4 refer to a specific location in the refrigeration cycle as specified by Fig. 1 and Fig. 2.

TABLE I

DECOUPLED I/O PAIRINGS

Controlled Input Variable	Controlled Output Variable
u_1 : EEV opening (%)	y_1 : Avg. System Pressure, $P_{(3+4)/2}$
u_2 : Compressor Speed (rpm),	y_2 : Differential Pressure, P_{3-4}
u_3 : Evaporator Fan Power (%)	y_3 : Avg. Evaporator Temp., $T_{(1+4)/2}$

The results of [1] demonstrated that a decentralized SISO control structure using this decoupled choice of control variables is more effective at tracking reference commands than a decentralized SISO control structure using the current choice of control variables used in industry. Moreover, results in [7] showed that the system, with the choice of variables from Table I, is sufficiently decoupled such that the decentralized SISO control approach is quantitatively as

Neera Jain (njain2@illinois.edu), Richard J. Otten (otten1@illinois.edu), and Andrew G. Alleyne (alleyne@illinois.edu) are with the Mechanical Science and Engineering Department at the University of Illinois at Urbana-Champaign, Urbana, IL 61801 USA (phone: 217-244-9993; fax: 217-244-6534).

effective as a MIMO control structure when the decoupled control variables are used.

Although the coupling in the system has been greatly reduced, it has not been eliminated. Feedforward control techniques have been applied to A/C & R systems to improve the regulation of evaporator superheat by reducing the effect of changes in other control inputs [4][8]. In this paper we will demonstrate the use of feedforward control in the decentralized framework to remove ‘disturbances’ in one output channel caused by the effect of coupling from other input channels in the system. Section II describes the generation of an identified model and an analysis of the plant. Section III describes the feedforward controller design, and results of the controller implementation on an experimental system are presented in Section IV.

II. SYSTEM IDENTIFICATION AND ANALYSIS

A. System Model Identification

In this paper, the dynamic response of an experimental A/C & R system is identified using a time domain system identification procedure. Three controllable inputs for a variable-speed A/C & R are considered: expansion valve opening, compressor speed, and evaporator fan speed (see u_1 through u_3 in Fig. 1). The condenser fan speed is not considered a controllable input because in many cases, such as automotive systems, the condenser air flow rate is a function of vehicle speed and acts as a disturbance to the feedback loop.

The output responses to random Gaussian combinations of all three inputs (see Fig. 3) around a nominal operating condition (see Table II) were collected on an A/C & R experimental test stand. For a more detailed description of the experimental system, see [8].

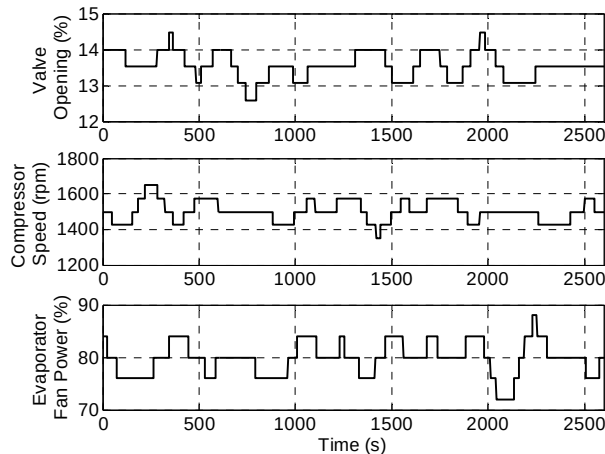


Fig. 3. Random Gaussian input signals for identification data.

A standard prediction error/maximum likelihood system identification algorithm in the System Identification Toolbox [9] was used to identify a third-order linear state space model using the I/O pairings described in Table I. The nominal operating value for each input and output parameter was subtracted from its measured signal, so all variables in

the model represent deviation variables. The signals were then scaled by their respective standard deviation to normalize the signals and improve the identification.

TABLE II
NOMINAL OPERATING CONDITION USED FOR SYSTEM ID

I/O Parameter	Parameter Description	Nominal Operating Condition
u_1	EEV Opening	14%
u_2	Compressor Speed	1500 rpm
u_3	Evaporator Fan Power	80%
y_1	Average Pressure	650.5 kPa
y_2	Differential Pressure	725 kPa
y_3	Avg. Evap. Refrigerant Temperature	8.5 °C

Fig. 4 shows the system identification results for the experimental system. Fig. 5 shows the cross-validation of the identified model. The complete state space representation of the identified model is included in the Appendix. The appropriate transformation matrices were applied such that the model represents the original unscaled system. Note that in Fig. 4, evaporator pressure, rather than average system pressure, was used for the identification. The model representation shown in the Appendix was transformed appropriately such that y_1 represents average system pressure.

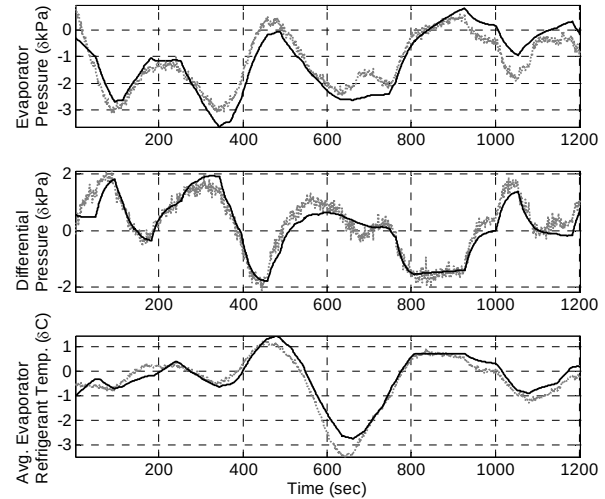


Fig. 4. Detrended system ID results using decoupled I/O pairings; y_1 : fit = 50.84%; y_2 : fit = 66.178%; y_3 : fit = 69.61%.

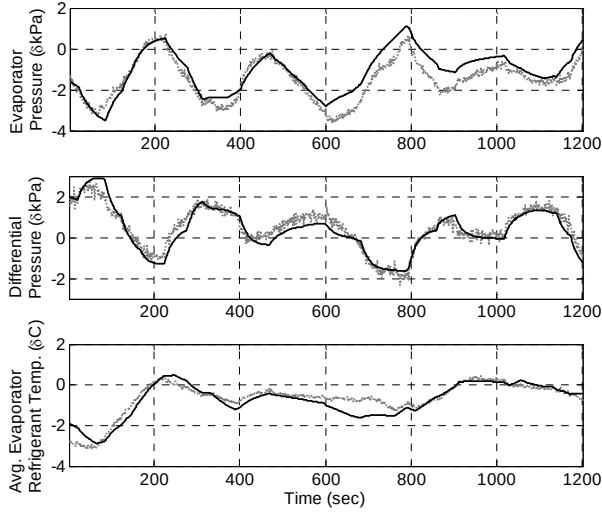


Fig. 5. Cross-validation of decoupled ID model; y_1 : fit = 43.53%; y_2 : fit = 65.78%; y_3 : fit = 56.5%.

B. Plant Analysis

Analysis given in [7] showed that the decoupled plant is indeed *generalized diagonally dominant* [10]. However, while the coupling in the system has been reduced, it has not been eliminated from the system altogether. The interactions between the inputs and outputs of a MIMO system are characterized by the off-diagonal elements of the plant transfer function matrix, $G(s)$, where

$$G(s) = \frac{Y(s)}{U(s)}. \quad (1)$$

The *scaled* plant transfer function matrix for our identified plant, $G(s)$ is given in the Appendix.

The desire to minimize root-mean-square (RMS) error and control effort is motivated by the need to minimize energy consumed during operation while maintaining tight temperature control. This leads us to evaluate the infinity norm of each of the transfer functions contained within $G(s)$ in order to identify where the strongest coupling still remains in the system. The infinity norm was evaluated over the system bandwidth ($\omega = 0$ to $\omega = 0.1$ rad/s); the results are shown in Equation (2).

$$\|G_{ij}(s)\|_{\infty}, i, j = \{1, 2, 3\} = \begin{bmatrix} 2.12 & 0.902 & 1.67 \\ 0.350 & 0.875 & 0.126 \\ 0.555 & 0.469 & 2.169 \end{bmatrix} \quad (2)$$

We are interested in the relative dominance of the response of each output to its respective input, over the response of each output to the two remaining inputs. For example, the infinity norm of the transfer function between u_1 and y_1 is 2.12, which is larger than the infinity norm of $\frac{y_1}{u_2}$ and $\frac{y_1}{u_3}$ by factors of 2.35 and 1.27, respectively. Similarly, the infinity norm of the transfer function between

u_2 and y_2 is 0.875, which is larger than the infinity norm of $\frac{y_2}{u_1}$ and $\frac{y_2}{u_3}$ by factors of 2.5 and 6.94, respectively.

Finally, when we examine y_3 , the average evaporator temperature, we find that infinity norm of the transfer function between u_3 and y_3 is larger than the infinity norm of $\frac{y_3}{u_1}$ and $\frac{y_3}{u_2}$ by factors of 3.91 and 4.63, respectively. This

is summarized in (3)

$$\begin{bmatrix} 2.12 & 0.902 & 1.67 \\ 0.350 & 0.875 & 0.126 \\ 0.555 & 0.469 & 2.169 \end{bmatrix} M = \begin{bmatrix} 1.00 & 0.426 & 0.788 \\ 0.400 & 1.00 & 0.144 \\ 0.256 & 0.216 & 1.00 \end{bmatrix} \quad (3)$$

where M is an appropriate transformation matrix, given in (4), that normalizes the diagonal elements of the input output norm to be unity. Equation (3) illustrates relative strengths of off-diagonal input output channels that still retain some level of coupling.

$$M = \begin{bmatrix} 0.374 & -0.249 & -0.0248 \\ 0.314 & 1.26 & 0.111 \\ -0.0457 & -0.109 & 0.444 \end{bmatrix} \quad (4)$$

For different systems and operating conditions, the particular magnitude of sensitivity to off-diagonal terms can vary. From time domain results presented in Fig. 6, we see that y_3 is sensitive to changes in both u_1 and u_2 .

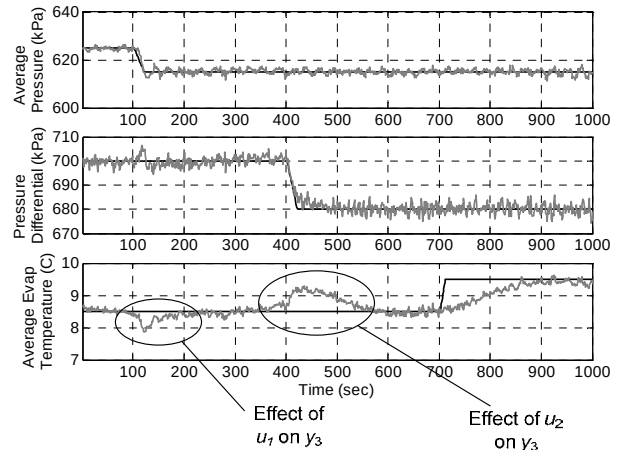


Fig. 6. Reference tracking results of a SISO decentralized controller.

For the sake of illustration, and without loss of generality, we examine the influence of u_2 on y_3 and the resulting benefits of a feedforward augmentation of that particular input-output channel. In a decentralized framework, we have the flexibility to introduce such a feedforward controller on a *single* control loop and to *tune it independently* of other control loops. This is a significant advantage over a MIMO control framework in which one would have to design a more complex feedforward controller even if the sensitivity is isolated to a single input-

output channel (as it is in this case).

III. CONTROLLER DESIGN

A. Decentralized SISO Feedback Controller

A decentralized feedback control approach using three SISO control loops, as shown in Fig. 7, is used to effectively track step references.

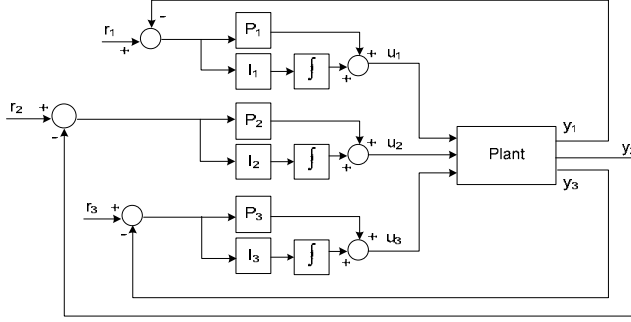


Fig. 7. Schematic describing SISO decentralized control framework.

A proportional-integral (PI) controller is used for each of the individual SISO control loops. The motivation for using this type of controller is that it is widely used in the HVAC industry. Table III shows the chosen gains for each of the PI controllers based on a procedure outlined in [7] to match a particular optimal control approach.

TABLE III PI GAINS FOR INDIVIDUAL SISO CONTROLLERS		
Input	Proportional Gain, P	Integral Gain, I
u_1 , EEV Opening	7.3232	0.1000
u_2 , Compressor Speed	3.5365	0.0816
u_3 , Evaporator Fan Speed	1.9423	0.0316

B. Feedforward Control Design

The influence of the coupling between y_3 and u_2 can be treated as a known disturbance on y_3 . Using the relationship between y_3 and u_2 identified in the system model given in the Appendix, a feedforward controller can be added to the feedback controller on y_3 (Fig. 8).

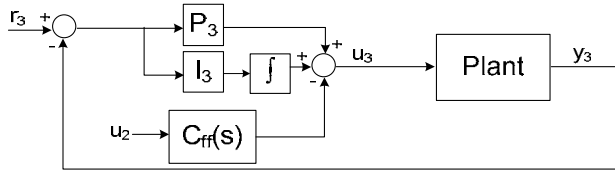


Fig. 8. Schematic describing feedforward plus feedback controller implementation for y_3 , average evaporator temperature.

The form of the feedforward controller is given in Equation (5), and the transfer function is given in Equation (6).

$$C_{ff}(s) = \frac{y_3(s)}{u_2(s)} \left(\frac{y_3(s)}{u_3(s)} \right)^{-1} \quad (5)$$

$$C_{ff}(s) = \frac{-0.0102s^2 - 0.000193s - 7.84 \times 10^{-7}}{0.0160s^2 + 0.000785 + 3.63 \times 10^{-6}} \quad (6)$$

IV. CONTROLLER IMPLEMENTATION

The feedforward controller on y_3 was implemented in simulation using the identified system model. Fig. 9 and Fig. 10 show the simulated system response without and with, respectively, the implementation of a feedforward controller on y_3 . Fig. 10 shows a significant reduction in sensitivity of y_3 to changes in r_2 (and consequently, u_2).

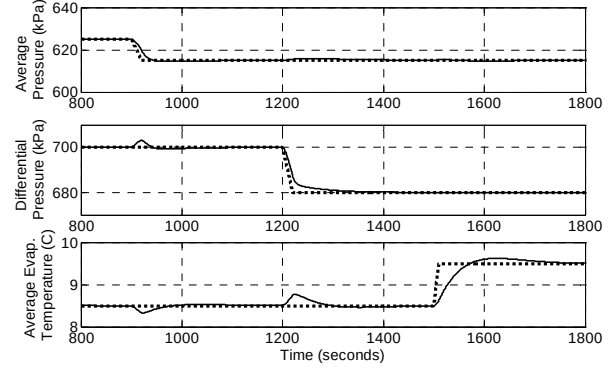


Fig. 9. Simulation results without feedforward compensation.

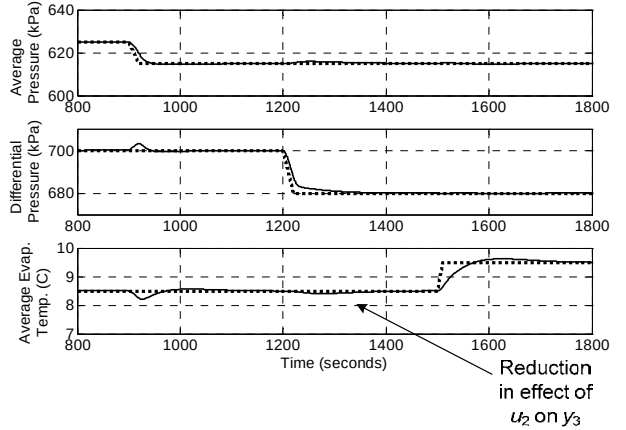


Fig. 10. Simulation results with feedforward compensation.

The feedforward controller was then implemented on the experimental A/C & R system used for the original model identification. Fig. 11 and Fig. 13 show the system response and actuation signals, respectively, *without* the addition of a feedforward controller. The effect of the feedforward controller combined with the decentralized feedback controller is shown in Fig. 12 and Fig. 14. Fig. 14 shows that u_3 , the evaporator fan power, decreases more sharply at $t = 400$ seconds than in Fig. 13 to compensate for the change in u_2 , compressor speed.

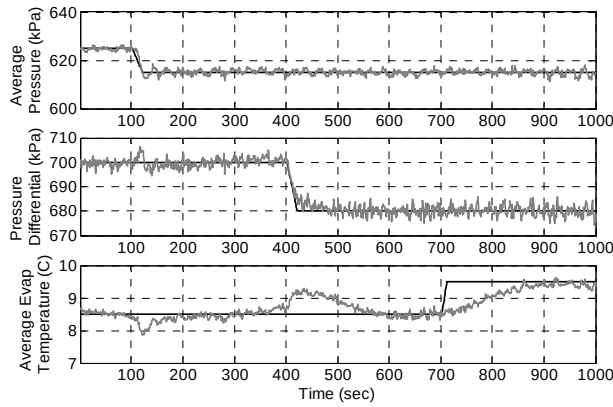


Fig. 11. Experimental results without feedforward compensation.

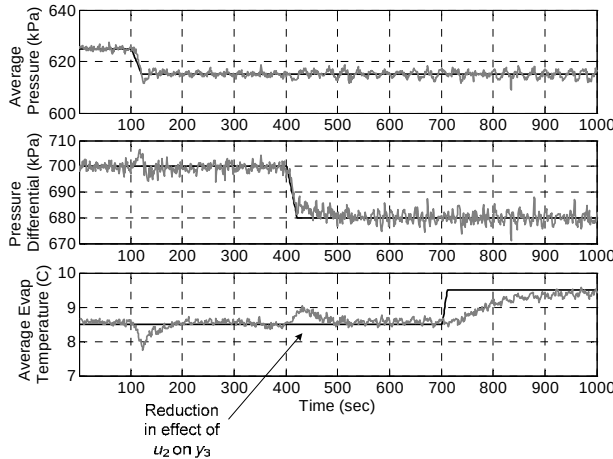


Fig. 12. Experimental results with feedforward compensation.

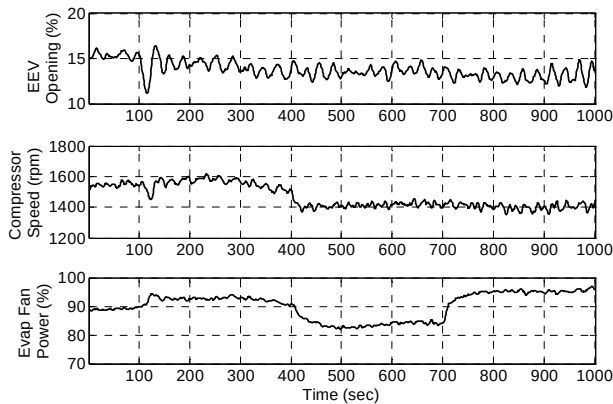


Fig. 13. Experimental actuation signals without feedforward compensation.

The maximum deviation and root-mean-square (RMS) error between y_3 , the average evaporator temperature signal, and its reference signal were evaluated over $t = 0$ to $t = 700$ seconds (see Table IV). The addition of the feedforward controller to the feedback controller on y_3 reduced the RMS error by 40%. When working with reference trajectories that have been optimized for efficient system operation, such a reduction in RMS error can translate into a reduction in energy consumption.

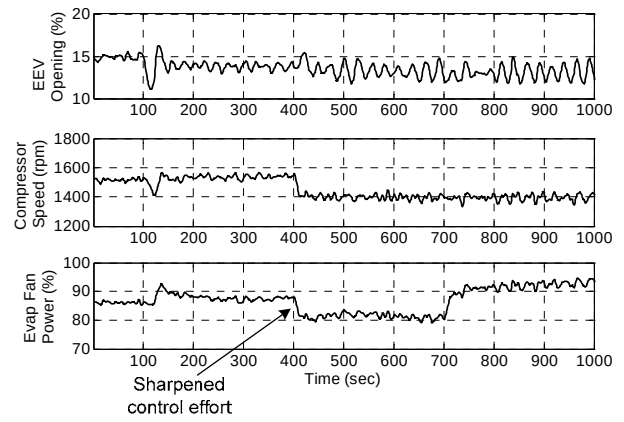


Fig. 14. Experimental actuation signals with feedforward compensation.

TABLE IV
COMPARISON BETWEEN FEEDBACK AND FEEDBACK + FEEDFORWARD COMPENSATION

Metric	Feedback Only	Feedback + Feedforward	Percent Change (%)
Maximum Deviation	0.789 °C	0.539 °C	-32
RMS Error	0.2689	0.1609	-40

Additionally, the advantage of implementing a feedforward controller in a decentralized control framework is highlighted by a comparison between the responses of y_1 , average system pressure, and y_2 , differential system pressure, in Fig. 11 and Fig. 12. The signals are nearly identical between the two figures, underscoring the fact that they are unaffected by the addition of a feedforward controller on y_3 .

V. CONCLUSION

This paper demonstrates that when using a decentralized control framework for a decoupled A/C & R system, the addition of a feedforward controller can improve the system response to reference signals and reduce the effect of system coupling. In a decentralized framework, the control engineer has the flexibility to perform this type of feedforward augmentation on a *single* control loop, a significant advantage over a MIMO control framework in which one would have to design a more complex feedforward controller even when system coupling is isolated to a single input-output channel. Furthermore, a reduction in the RMS error of the controlled signal was shown when feedforward compensation was implemented. For a reference trajectory that has been optimized for efficient system operation, a reduction in RMS error can translate into a reduction in energy consumption. Moreover, minimizing deviation from the optimized reference trajectory will ensure safe operation of components, particularly the compressor.

Future work will examine strategies for mitigation of the uncertainty associated with the feedforward controller. This includes adaptive feedforward algorithms [11] that use input-output system behavior to identify the appropriate feedforward transfer function, $C_{ff}(s)$ in Equation (6). A

mapping or scheduling of feedforward functions, such as in [8], based on operating conditions is also a topic for future work.

APPENDIX

The (unscaled) identified state space $[A, B, C, D]$ system model is given in (7).

$$\begin{aligned} A &= \begin{bmatrix} -0.00441 & 0.000559 & -0.000461 \\ -0.00336 & -0.00684 & 0.000858 \\ -0.00154 & -0.0201 & -0.0490 \end{bmatrix} \\ B &= \begin{bmatrix} 0.000121 & -9.46 \times 10^{-8} & -0.000137 \\ -0.000415 & 2.40 \times 10^{-6} & -7.50 \times 10^{-5} \\ -0.00178 & 7.14 \times 10^{-5} & -0.000160 \end{bmatrix} \\ C &= \begin{bmatrix} 105 & -284 & -47.3 \\ -18.3 & 3.95 & 127 \\ -21.7 & -15.3 & -2.46 \end{bmatrix} \quad D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \\ \delta \mathbf{u} &= \begin{bmatrix} a_v & \omega_c & p_f \end{bmatrix}^T \quad \delta \mathbf{y} = \begin{bmatrix} P_{(3+4)/2} & P_{3-4} & T_{(1+4)/2} \end{bmatrix} \end{aligned} \quad (7)$$

The *scaled* plant transfer function matrix, $G_{scaled}(s)$, is given in (8).

ACKNOWLEDGEMENT

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$$G_{scaled}(s) = \begin{bmatrix} \frac{0.0113s^2 + 0.000722s + 3.55 \times 10^{-6}}{s^3 + 0.0603s^2 + 0.000600s + 1.67 \times 10^{-6}} & \frac{-0.00121s^2 - 0.000306s - 1.51 \times 10^{-6}}{s^3 + 0.0603s^2 + 0.000600s + 1.67 \times 10^{-6}} & \frac{0.00292s^2 + 0.000129s - 2.79 \times 10^{-6}}{s^3 + 0.0603s^2 + 0.000600s + 1.67 \times 10^{-6}} \\ \frac{-0.019s^2 - 0.000144s - 2.24 \times 10^{-7}}{s^3 + 0.0603s^2 + 0.000600s + 1.67 \times 10^{-6}} & \frac{0.0453s^2 + 0.000486s + 1.34 \times 10^{-6}}{s^3 + 0.0603s^2 + 0.000600s + 1.67 \times 10^{-6}} & \frac{-0.00658s^2 + (4.13 \times 10^{-5})s + 5.67 \times 10^{-8}}{s^3 + 0.0603s^2 + 0.000600s + 1.67 \times 10^{-6}} \\ \frac{0.00653s^2 + 0.000192s + 9.29 \times 10^{-7}}{s^3 + 0.0603s^2 + 0.000600s + 1.67 \times 10^{-6}} & \frac{-0.0102s^2 - 0.000193s - 7.84 \times 10^{-7}}{s^3 + 0.0603s^2 + 0.000600s + 1.67 \times 10^{-6}} & \frac{0.0160s^2 + 0.000785s + 3.63 \times 10^{-6}}{s^3 + 0.0603s^2 + 0.000600s + 1.67 \times 10^{-6}} \end{bmatrix} \quad (8)$$